

Egocentric 3D Pose Estimation with Temporally Guided Pose Refinement

Seongmin Baek
Electronics and Telecommunications
Research Institute (ETRI)
Daejeon, Republic of Korea
baeksm@etri.re.kr

Youn-Hee Gil
Electronics and Telecommunications
Research Institute (ETRI)
Daejeon, Republic of Korea
yhgil@etri.re.kr

Hyunjun Lim
Electronics and Telecommunications
Research Institute (ETRI)
Daejeon, Republic of Korea
tp02134@kaist.ac.kr

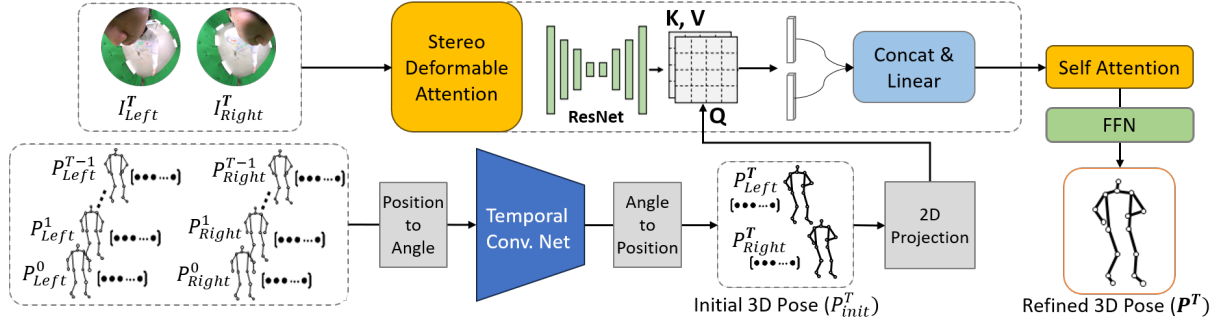


Figure 1: a causal TCN over the previous T frames, represented as head-centered joint rotations (ZYX Euler), predicts a motion-conditioned pose prior. The prior is mapped back to 3D joints and reprojected onto left/right egocentric images to form 2D keypoint queries (Q). Stereo RGBs are encoded with a ResNet to produce feature maps used as keys/values (K, V). A single-layer Transformer refines the pose via cross- and self-attention; an FFN head outputs the final 3D pose.

Abstract

Unlike single-frame pose estimation, human motion is inherently temporally continuous, and this challenge is further compounded by egocentric views, which suffer from severe self-occlusions and limited FoV, making per-frame inference unstable. We address these issues with a two-stage pipeline: a causal Temporal Convolutional Network (TCN) over the previous T frames predicts a motion-conditioned pose prior, and a single-layer Transformer refines this prior with current egocentric visual features to produce the final 3D pose. This design yields accurate and temporally coherent egocentric pose estimation with minimal latency.

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1 Introduction

Egocentric 3D human pose estimation is increasingly important with the rise of wearable devices (e.g., HMDs, AR glasses), yet

remains challenging due to limited field of view (FoV), frequent self-occlusions, and head-induced camera motion.

Human motion is continuous and temporally structured, and recent methods inject temporal reasoning into pose pipelines. PoseFormer [Zheng et al. 2021] models spatial and temporal relations but examines both past and future context, making real-time application difficult. VideoPose3D [Pavlo et al. 2019] uses temporal convolutions over 2D keypoints but discards image cues and assumes accurate 2D inputs. UnrealEgo [Akada et al. 2022] provides a stereo RGB egocentric dataset with a dual-fisheye rig and a single-frame baseline model, and EgoPoseFormer [Yang et al. 2024] improves accuracy via multi-layer refinement at the cost of increased latency.

We propose a lightweight framework for egocentric stereo 3D pose estimation that avoids offline temporal reasoning and heavy multi-layer refinement. A causal TCN over the previous T frames generates a motion-conditioned 3D pose prior, which is then refined by a single-layer Transformer using current stereo image cues, preserving temporal coherence with minimal latency.

Contributions.

- A lightweight two-stage pipeline for egocentric stereo 3D pose estimation, exploiting temporal continuity via a causal TCN to generate a motion-conditioned pose prior.
- A single-layer Transformer refiner that achieves accuracy comparable to iterative multi-layer refinement while reducing parameters and latency.

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2 Method

We propose a two-stage pipeline for egocentric 3D human pose estimation that leverages temporal continuity and a lightweight spatial refiner.

2.1 Temporal Pose Prediction via TCN

We employ a causal Temporal Convolutional Network (TCN) with a sliding window of length T (here $T=3$) and stride 1. The TCN predicts the current pose from the preceding T frames, enforcing temporal consistency.

To mitigate head-induced camera motion in egocentric coordinates, we represent poses as joint rotations using the head joint as the kinematic root (ZYX Euler in our implementation) and recover 3D joint positions via differentiable forward kinematics. This head-anchored rotation parameterization is invariant to global head motion and better captures relative dynamics.

2.2 Transformer-Based Pose Refinement

While the causal TCN prior enforces temporal consistency, it may lack fine-grained spatial accuracy due to occlusions and limited FoV in egocentric views. To recover precise spatial joint structures, the predicted 3D joints are projected onto the stereo images to obtain 2D keypoints, which serve as the queries (Q). Stereo RGBs are processed by a ResNet backbone to extract feature maps used as keys/values (K, V). Unlike EgoPoseFormer [Yang et al. 2024], which performs iterative multi-layer refinement, we apply a single-layer Transformer refiner. Because the TCN prior is already close to the target, one layer suffices to retain accuracy while reducing computation and latency.

Training objective (shared by both stages). We use MPJPE for 3D joints and MSE for 2D reprojection. Let $\hat{\mathbf{X}}_j \in \mathbb{R}^3$ be the predicted 3D joint j (mm; head-centered) and $\mathbf{X}_j$ the ground truth. For view $v \in \{L, R\}$, let $\Pi_v(\cdot)$ be the differentiable projection with calibrated intrinsics/extrinsics, and $\hat{\mathbf{u}}_j^{(v)} = \Pi_v(\hat{\mathbf{X}}_j)$, $\mathbf{u}_j^{(v)}$ the 2D keypoints (in the training image coordinate system).

$$\mathcal{L}_{3D} = \frac{1}{J} \sum_{j=1}^J \|\hat{\mathbf{X}}_j - \mathbf{X}_j\|_2 \quad (\text{MPJPE, mm}), \quad (1)$$

$$\mathcal{L}_{2D}^{(v)} = \frac{1}{J} \sum_{j=1}^J \|\hat{\mathbf{u}}_j^{(v)} - \mathbf{u}_j^{(v)}\|_2^2 \quad (\text{MSE}), \quad (2)$$

$$\mathcal{L} = \lambda_{3D} \mathcal{L}_{3D} + \lambda_{2D} \sum_{v \in \{L, R\}} \mathcal{L}_{2D}^{(v)}. \quad (3)$$

We apply the same objective to both stages:

$$\mathcal{L}_{\text{TCN}} = \mathcal{L}, \quad \mathcal{L}_{\text{TR}} = \mathcal{L}, \quad \mathcal{L}_{\text{total}} = \lambda_{\text{TCN}} \mathcal{L}_{\text{TCN}} + \lambda_{\text{TR}} \mathcal{L}_{\text{TR}} \quad (4)$$

3 Experiment

We evaluate our method on the UnrealEgo-RW dataset(83K), a real-world egocentric stereo dataset for 3D human pose estimation. The model is trained on 68K samples and evaluated on a separate test set of 10K samples. We compare our method with two baselines: the UnrealEgo baseline [Akada et al. 2022] and the EgoPoseFormer model [Yang et al. 2024]. All methods are evaluated within the same head-mounted coordinate frame provided by UnrealEgo-RW to

ensure a fair comparison. Qualitative results are shown in Fig.2, and quantitative comparisons are presented in Table1. Metrics include MPJPE and PA-MPJPE (mm, head-centered for MPJPE).

To assess component contributions, we conduct an ablation between (i) TCN only and (ii) TCN + single-layer Transformer (ours). As shown in Table 1, adding the lightweight Transformer refiner improves accuracy over the TCN prior while preserving temporal coherence, while requiring fewer parameters and achieving faster inference speed (fps, measured with batch size = 1 on Intel i9-3.0GHz and RTX 4090).

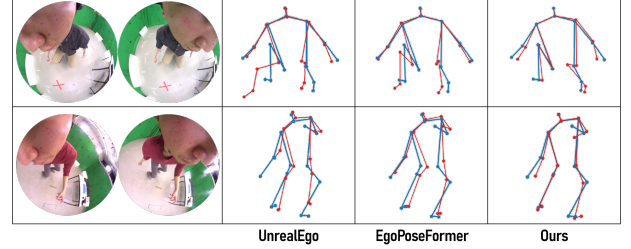


Figure 2: Qualitative visualization on UnrealEgo-RW. 3D pose predictions and ground truth are shown in red and blue, respectively.

Table 1: Quantitative results on UnrealEgo-RW [Akada et al. 2024] with mm-scale.

Method	MPJPE	PA-MPJPE	FPS	Param(M)
UnrealEgo	61.24	51.45	17.05	107.30
EgoPoseFormer	40.58	36.29	6.02	12.967
TCN ₃	35.15	29.87	99.9	0.054
TCN ₅	35.44	30.89	97.1	0.054
Ours(TCN ₃ +PRFormer)	31.20	27.99	15.06	12.235

4 Conclusion

This paper presents a lightweight egocentric 3D human pose estimation framework that employs a transformer-based spatial refiner guided by temporally predicted initial poses. Compared with iterative refinement (e.g., EgoPoseFormer [Yang et al. 2024]), our method reduces computational cost while maintaining accuracy, as the initial prediction is already sufficiently precise. Experiments on the UnrealEgo-RW dataset demonstrate the effectiveness of our approach in egocentric scenarios.

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